Analysis of injury and physical fitness using factor analysis

Code

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Data and Results (injury and physical performance test)

Summary of injury data

	All cases					During the season			
Year	Athlete	Issues	Illness	Acute injury	Overuse injury	Issues	Illness	Acute injury	Overuse injury
2020	163	255	40	117	100	183	37	92	54
2021	107	259	49	108	104	122	37	47	38
2023	140	328	72	140	118	259	68	112	81
2024	85	157	32	71	56	129	29	61	39

	Distinct	Average of weekly		
Year	Athlete	Timeloss	Match hr	Training hr
2020	163	113.50	89.41	690.56
2021	107	56.61	47.12	390.07
2023	140	64.12	60.84	760.43
2024	85	23.53	21.97	367.13

2020

Year: 2020	Missing	Mean ± SD	Median (IQR)
age	0/163	21.92 ± 3.97	21.40 (4.95)
leg_length_malleolus	163/163 (100.00%)	–	–
height_cm	41/163 (25.15%)	169.08 ± 5.84	169.00 (8.00)
weight_fp	49/163 (30.06%)	63.72 ± 6.43	64.00 (7.75)
sprint_10m	163/163 (100.00%)	–	–
sprint_20m	63/163 (38.65%)	3.12 ± 0.11	3.10 (0.15)
sprint_30m	63/163 (38.65%)	4.44 ± 0.16	4.41 (0.21)
sprint_40m	63/163 (38.65%)	5.74 ± 0.22	5.71 (0.31)
agility_left	69/163 (42.33%)	10.24 ± 0.29	10.17 (0.42)
agility_right	66/163 (40.49%)	10.18 ± 0.31	10.16 (0.44)
cmj_total	35/163 (21.47%)	33.38 ± 4.52	33.00 (5.50)
cmj_left	163/163 (100.00%)	–	–
cmj_right	163/163 (100.00%)	–	–
ankle_jump_left	47/163 (28.83%)	1.17 ± 0.16	1.16 (0.22)

Year: 2020	Missing	Mean ± SD	Median (IQR)
ankle_jump_right	47/163 (28.83%)	1.19 ± 0.15	1.18 (0.18)
side_jump_left	137/163 (84.05%)	72.31 ± 5.92	73.50 (4.00)
side_jump_right	137/163 (84.05%)	73.35 ± 4.38	73.50 (7.00)
max_power_left	34/163 (20.86%)	556.19 ± 75.53	552.00 (98.00)
max_power_right	34/163 (20.86%)	550.11 ± 77.65	541.00 (101.00)
max_force_left	36/163 (22.09%)	1,064.36 ± 157.58	1,033.00 (212.50)
max_force_right	36/163 (22.09%)	1,068.44 ± 162.99	1,050.00 (215.00)
hamstring_left	36/163 (22.09%)	335.18 ± 48.26	336.00 (56.50)
hamstring_right	36/163 (22.09%)	348.99 ± 46.96	350.00 (59.50)
extra_weight	163/163 (100.00%)	–	–
hip_squeeze_0_left	34/163 (20.86%)	157.98 ± 24.96	155.00 (30.00)
hip_squeeze_0_right	34/163 (20.86%)	161.57 ± 23.91	159.00 (30.00)
hip_pull_0_left	34/163 (20.86%)	146.49 ± 20.31	146.00 (28.00)
hip_pull_0_right	34/163 (20.86%)	153.52 ± 23.37	151.00 (33.00)

2021

Year: 2021	Missing	Mean ± SD	Median (IQR)
age	0/107	21.88 ± 3.84	20.89 (5.07)
leg_length_malleolus	0/107	0.88 ± 0.04	0.89 (0.05)
height_cm	18/107 (16.82%)	170.11 ± 5.62	170.00 (7.00)
weight_fp	0/107	65.77 ± 6.36	65.80 (8.74)
sprint_10m	107/107 (100.00%)	–	–
sprint_20m	34/107 (31.78%)	3.16 ± 0.13	3.15 (0.20)
sprint_30m	35/107 (32.71%)	4.47 ± 0.17	4.48 (0.25)
sprint_40m	35/107 (32.71%)	5.79 ± 0.26	5.80 (0.28)
agility_left	41/107 (38.32%)	10.18 ± 0.31	10.16 (0.41)
agility_right	39/107 (36.45%)	10.17 ± 0.28	10.14 (0.39)
cmj_total	9/107 (8.41%)	33.47 ± 4.84	33.40 (6.08)
cmj_left	107/107 (100.00%)	–	–
cmj_right	107/107 (100.00%)	–	–
ankle_jump_left	19/107 (17.76%)	1.19 ± 0.16	1.18 (0.18)
ankle_jump_right	19/107 (17.76%)	1.22 ± 0.15	1.20 (0.22)
side_jump_left	97/107 (90.65%)	63.70 ± 5.25	65.00 (8.00)
side_jump_right	97/107 (90.65%)	66.30 ± 4.22	66.50 (5.50)
max_power_left	3/107 (2.80%)	543.14 ± 88.22	536.50 (123.50)
max_power_right	3/107 (2.80%)	539.42 ± 92.30	527.00 (111.50)
max_force_left	3/107 (2.80%)	998.02 ± 145.63	978.50 (231.36)
max_force_right	3/107 (2.80%)	1,003.78 ± 150.18	979.50 (223.50)
hamstring_left	9/107 (8.41%)	325.54 ± 45.38	324.00 (53.00)
hamstring_right	9/107 (8.41%)	337.87 ± 50.48	340.00 (61.75)

Year: 2021	Missing	Mean ± SD	Median (IQR)
extra_weight	107/107 (100.00%)	–	–
hip_squeeze_0_left	3/107 (2.80%)	158.29 ± 26.48	156.00 (31.50)
hip_squeeze_0_right	3/107 (2.80%)	163.87 ± 25.27	161.00 (30.50)
hip_pull_0_left	3/107 (2.80%)	153.83 ± 23.44	150.50 (33.00)
hip_pull_0_right	3/107 (2.80%)	160.51 ± 24.16	158.50 (30.25)

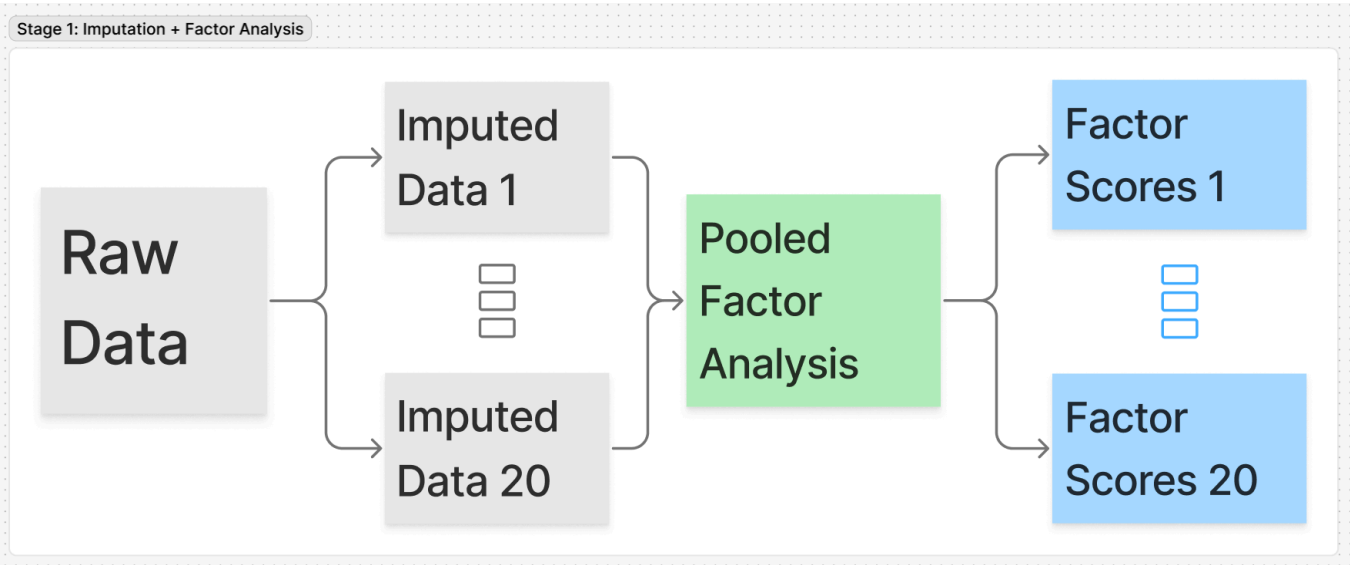
2023

Year: 2023	Missing	Mean ± SD	Median (IQR)
age	0/140	22.54 ± 3.73	22.54 (4.96)
leg_length_malleolus	6/140 (4.29%)	0.90 ± 0.04	0.90 (0.06)
height_cm	127/140 (90.71%)	170.54 ± 6.80	173.00 (10.00)
weight_fp	11/140 (7.86%)	65.90 ± 7.58	65.40 (9.60)
sprint_10m	61/140 (43.57%)	1.71 ± 0.08	1.71 (0.09)
sprint_20m	61/140 (43.57%)	3.15 ± 0.13	3.14 (0.18)
sprint_30m	61/140 (43.57%)	4.45 ± 0.18	4.41 (0.25)
sprint_40m	62/140 (44.29%)	5.75 ± 0.24	5.69 (0.33)
agility_left	75/140 (53.57%)	9.92 ± 0.33	9.89 (0.31)
agility_right	78/140 (55.71%)	10.04 ± 0.33	10.00 (0.33)
cmj_total	36/140 (25.71%)	32.13 ± 4.42	31.95 (5.67)
cmj_left	39/140 (27.86%)	22.12 ± 3.65	22.60 (4.60)
cmj_right	42/140 (30.00%)	22.30 ± 4.33	22.00 (5.15)
ankle_jump_left	49/140 (35.00%)	1.49 ± 0.19	1.50 (0.25)
ankle_jump_right	53/140 (37.86%)	1.48 ± 0.22	1.46 (0.32)
side_jump_left	137/140 (97.86%)	63.67 ± 4.16	65.00 (4.00)
side_jump_right	137/140 (97.86%)	62.67 ± 6.66	61.00 (6.50)
max_power_left	34/140 (24.29%)	558.24 ± 87.97	554.50 (113.50)
max_power_right	34/140 (24.29%)	560.13 ± 89.98	558.00 (122.00)
max_force_left	34/140 (24.29%)	1,042.75 ± 147.13	1,056.50 (200.25)
max_force_right	34/140 (24.29%)	1,053.39 ± 146.17	1,062.00 (196.25)
hamstring_left	23/140 (16.43%)	327.24 ± 47.27	323.00 (57.00)
hamstring_right	23/140 (16.43%)	337.46 ± 51.21	341.00 (72.00)
extra_weight	140/140 (100.00%)	–	–
hip_squeeze_0_left	11/140 (7.86%)	169.98 ± 29.45	169.00 (37.00)
hip_squeeze_0_right	11/140 (7.86%)	171.60 ± 29.48	169.00 (37.00)
hip_pull_0_left	11/140 (7.86%)	151.74 ± 23.61	150.00 (31.00)
hip_pull_0_right	11/140 (7.86%)	152.99 ± 22.73	151.00 (28.00)

2024

Year: 2024	Missing	Mean ± SD	Median (IQR)
age	0/85	22.56 ± 4.10	21.73 (5.83)
leg_length_malleolus	5/85 (5.88%)	0.90 ± 0.04	0.90 (0.06)
height_cm	69/85 (81.18%)	169.62 ± 5.56	168.50 (8.00)
weight_fp	2/85 (2.35%)	65.68 ± 5.75	65.20 (6.90)
sprint_10m	24/85 (28.24%)	1.78 ± 0.07	1.78 (0.11)
sprint_20m	24/85 (28.24%)	3.17 ± 0.12	3.16 (0.17)
sprint_30m	24/85 (28.24%)	4.47 ± 0.17	4.47 (0.25)
sprint_40m	24/85 (28.24%)	5.78 ± 0.23	5.74 (0.31)
agility_left	39/85 (45.88%)	9.95 ± 0.29	9.89 (0.34)
agility_right	41/85 (48.24%)	10.00 ± 0.32	9.96 (0.46)
cmj_total	7/85 (8.24%)	31.07 ± 3.61	30.20 (4.88)
cmj_left	10/85 (11.76%)	21.77 ± 3.88	21.60 (4.95)
cmj_right	9/85 (10.59%)	21.60 ± 3.81	21.50 (5.93)
ankle_jump_left	16/85 (18.82%)	1.49 ± 0.16	1.49 (0.23)
ankle_jump_right	16/85 (18.82%)	1.48 ± 0.18	1.47 (0.25)
side_jump_left	85/85 (100.00%)	–	–
side_jump_right	85/85 (100.00%)	–	–
max_power_left	1/85 (1.18%)	545.62 ± 75.88	548.00 (91.00)
max_power_right	1/85 (1.18%)	543.51 ± 77.01	545.00 (100.25)
max_force_left	1/85 (1.18%)	1,010.24 ± 137.65	1,016.50 (171.50)
max_force_right	1/85 (1.18%)	1,011.39 ± 133.25	1,027.00 (163.00)
hamstring_left	11/85 (12.94%)	316.57 ± 41.68	310.00 (53.00)
hamstring_right	11/85 (12.94%)	335.80 ± 48.48	345.00 (62.50)
extra_weight	85/85 (100.00%)	–	–
hip_squeeze_0_left	1/85 (1.18%)	173.11 ± 32.04	170.50 (30.00)
hip_squeeze_0_right	1/85 (1.18%)	177.37 ± 32.77	173.50 (38.00)
hip_pull_0_left	1/85 (1.18%)	157.44 ± 27.98	157.50 (27.25)
hip_pull_0_right	1/85 (1.18%)	160.30 ± 28.32	160.00 (29.25)

Missing test data and multiple imputation



Multiple Imputation

- Multiple imputation gives multiple version (in our case, I have used 20) of complete dataset imputed based on other variables which are not missing.
- Only physical performance test (PPT) variables with no more than 60% missing were used in each years were used for the analysis (the variables in red of above table were not used)
- Only using the selected PPT variables, a pooled factor analysis was performed and scores were calculated for each athlete in all 20 imputed datasets.
- The scores were used as continuous measure of underlying factor (for example physical performance) and were also categorized into 3 equal parts (using quantiles) and labelled as high, medium, low.

Factor Analysis

What factor analysis does?

Factor Analysis:

- is a method used to group related physical tests into a smaller number of meaningful factors
- identifies underlying abilities (e.g. speed, strength, hip function) that explain why some tests are correlated
- helps summarize complex test batteries into clear, interpretable scores

Why factor analysis fits this study

- Many of the physical tests measure overlapping aspects of performance
- Treating each test variables separately would repeat the same information and over-emphasize sprint measures
- Here, factor analysis:
  - Reduces repetition of similar tests
  - Identifies distinct physical capacities
  - Estimate scores that better reflect true performance in identified factor domain

One vs multiple factors in factor analysis

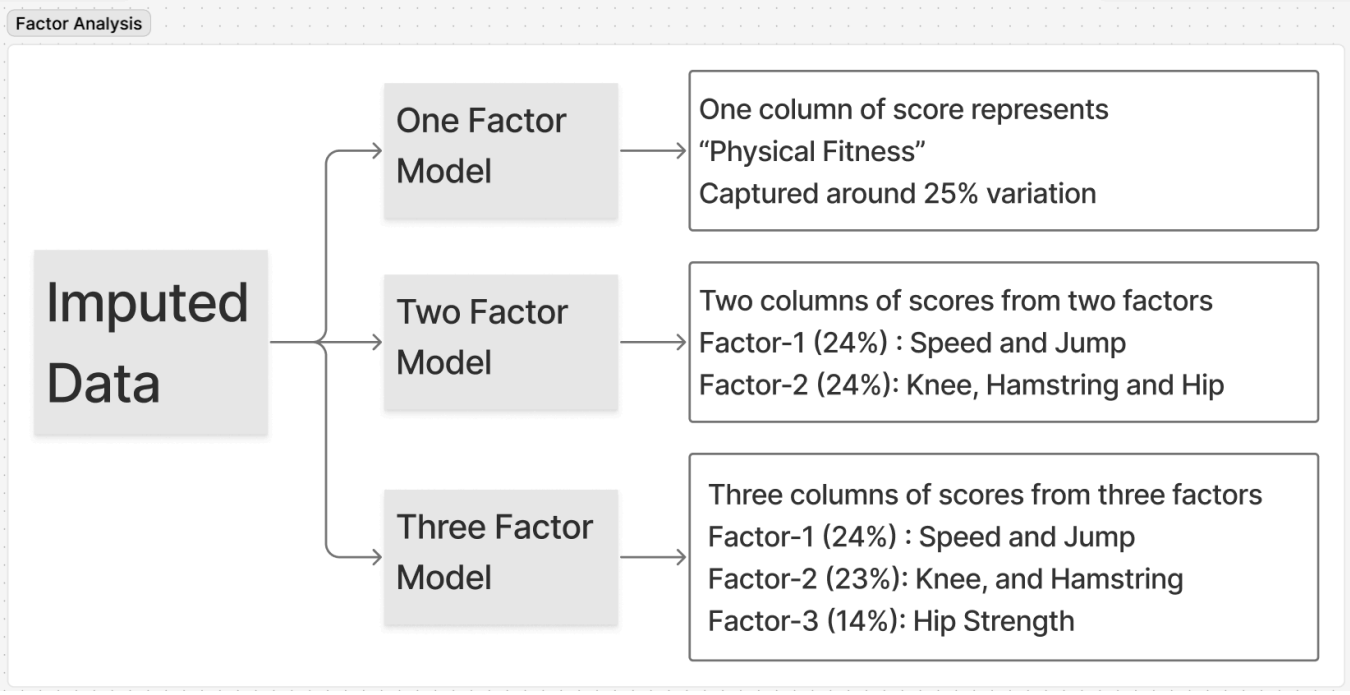
One-factor model

- Factor analysis can be used to identify **one or more underlying factors** from test data
- These factors are identified by using the **shared information across all tests**
- If the data reflect **one true underlying** ability:
  - A **1-factor model** may be sufficient
  - This single factor will typically show **strong correlations with most test variables**
  - Example: a general **overall physical performance** score
- If the data reflect **more than one underlying ability**:
  - A **1-factor model may not capture all available information**
  - Important differences between types of performance may be lost

Multiple factor model

- Using **more than one factor** allows the data to:
  - Separate **different physical performance domains**
  - Represent abilities that are related but not identical
- Example:
  - One factor may reflect being **good in speed and jumping tasks**
  - Another factor may reflect being **good in force and power tasks**
  - An athlete may perform well in one domain but not the other which leads to different types of injury risk
- In this case:
  - Multiple-factor models provide a **more detailed meaningful description of performance**
  - The factors identified may differ from the single factor obtained from a 1-factor model

Factor analysis with 1-3 factors model



Factor Analysis

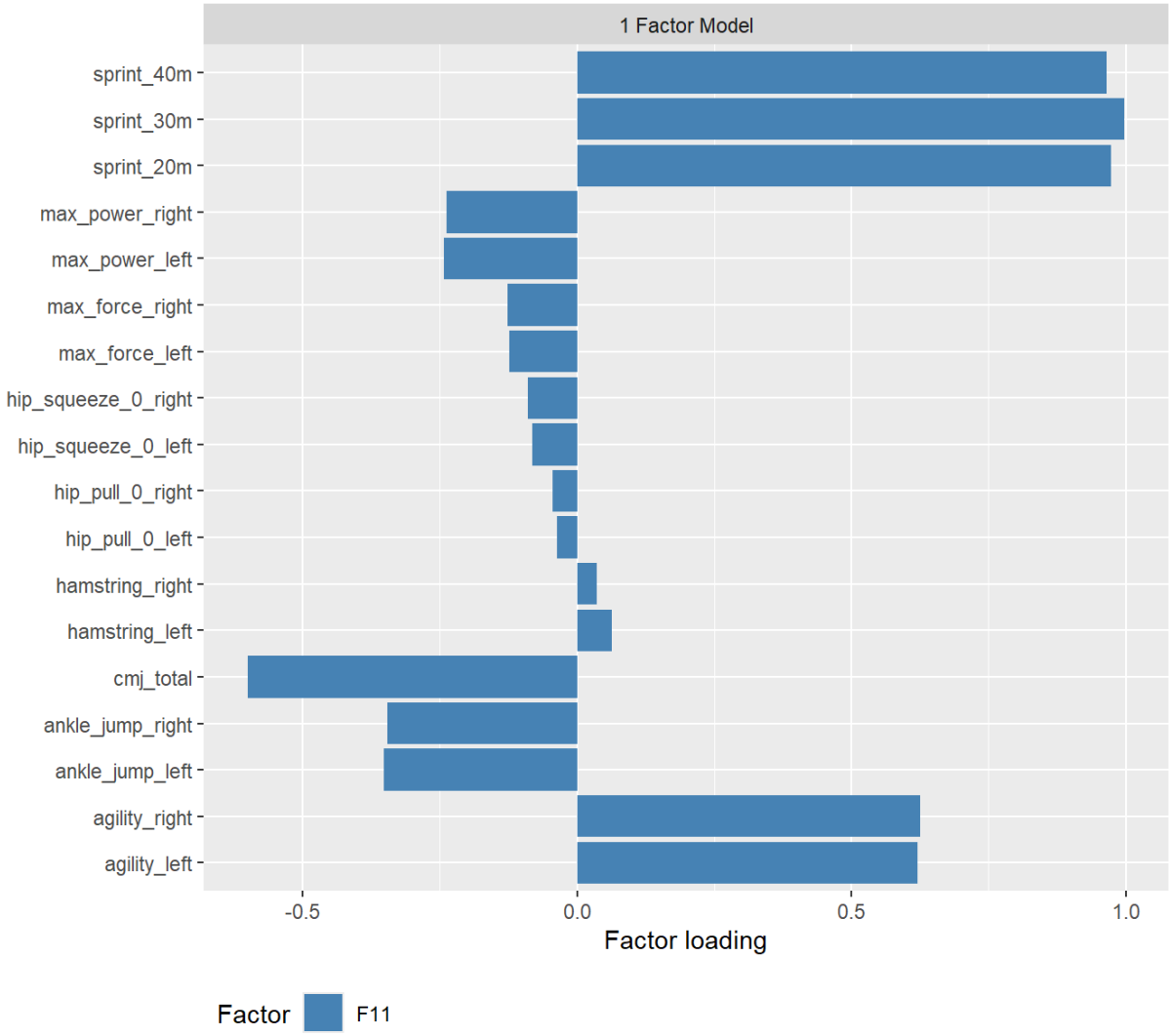
Factor and Factor Loadings

Loading are weights that indicate how much each test variable contributes to the factor. - A test variable with a high loading on a factor is strongly associated with that factor and contributes more to the factor

score. - A test variable with a low loading on a factor is weakly associated with that factor and contributes less to the factor score. - The factor score for each athlete is calculated by multiplying their test variable values by the corresponding factor loadings and summing these products across all test variables.

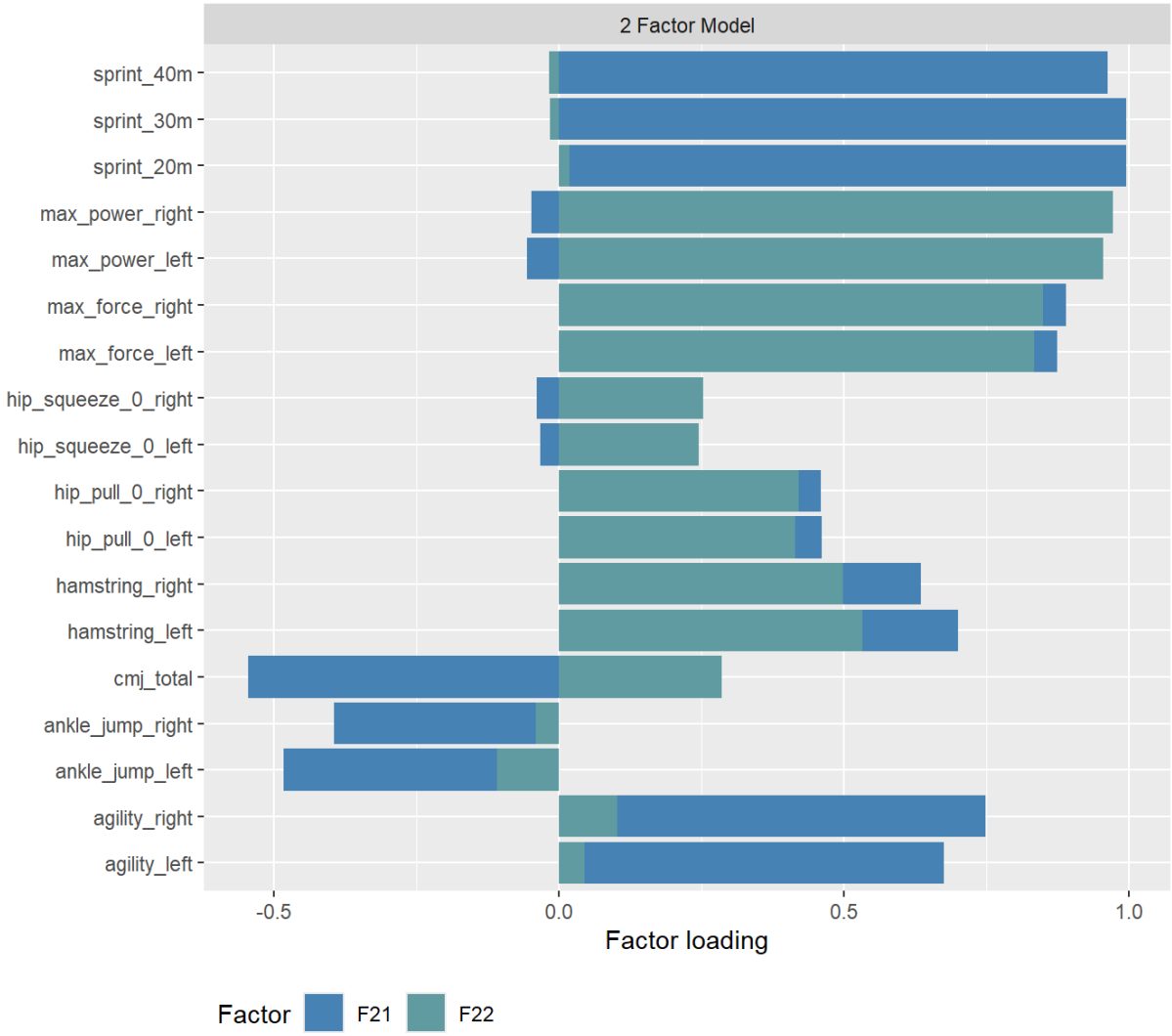
Plots

1 Factor Model



In the 1-factor model, all test variables have moderate to high loadings on the single factor, indicating that they all contribute to a this one factor (lets call it general physical performance factor). Although all variables have some contribution to the factor, sprint and jump related variables have the highest loadings, suggesting that the factor is most strongly associated with sprinting and jumping performance.

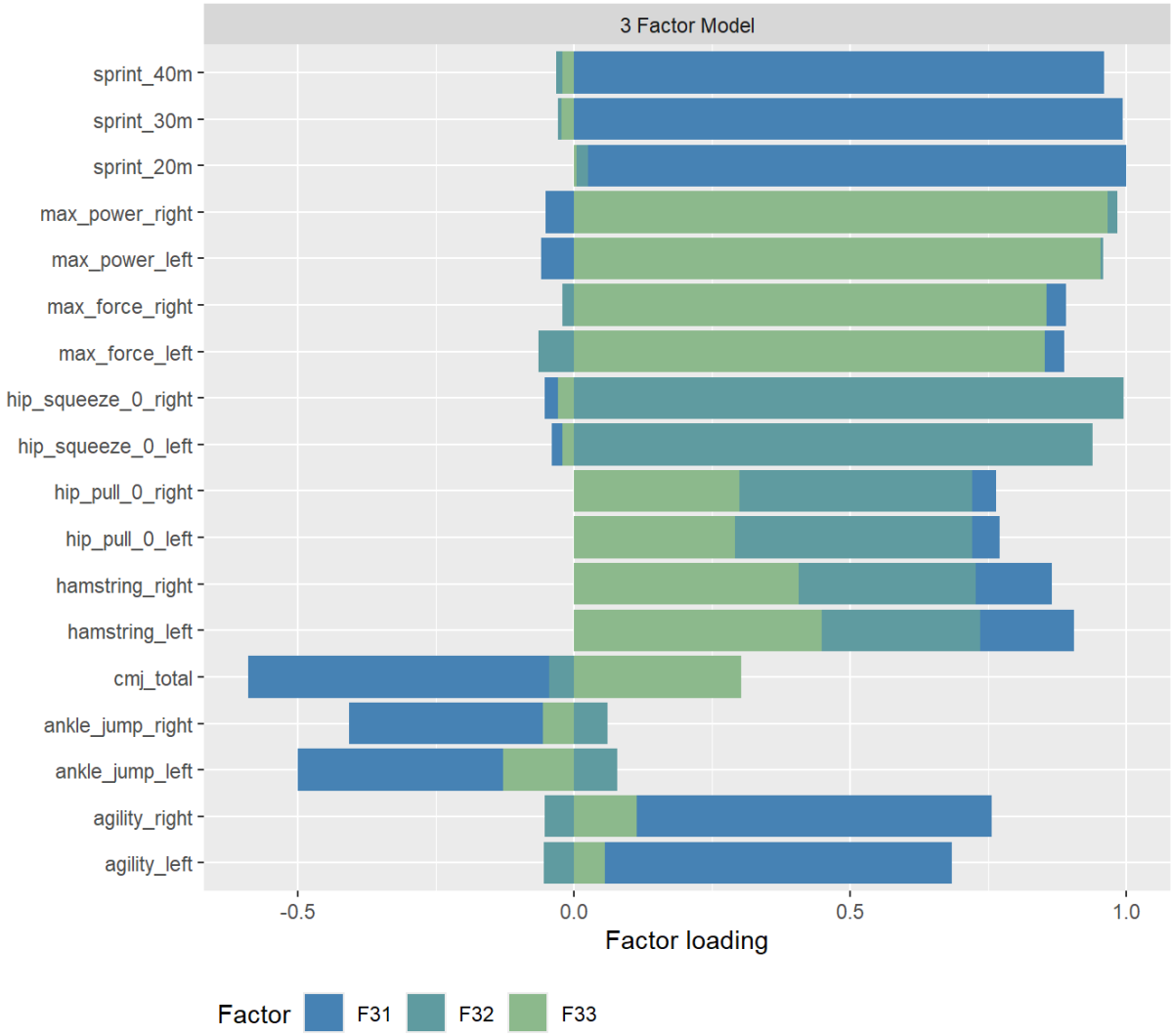
2 Factor Model



In the 2-factor model, the first factor has high loadings on sprint and jump related variables, while the second factor has high loadings on force and power related variables. This suggests that the first factor may represent a speed and power domain of physical performance, while the second factor may represent a strength and force domain.

When using 2-factor model and using scores from second factor, we can identify athletes who are good in force and power related tasks but not in sprint and jump related tasks which may be missed if we use scores from 1-factor model.

3 Factor Model



In the 3-factor model, we captured yet another domain of physical performance related to hip function. In this model, the first factor has high loadings on sprint and jump related variables, the second factor has high loadings on hip squeeze and pull related variables, which in the new dimension of physical performance related to hip function, and the third factor has high loadings on force and power related variables. This suggests that the first factor may represent a speed and power domain of physical performance, the second factor may represent a hip function domain, and the third factor may represent a strength and force domain.

When using 3-factor model, we can identify athletes who are good in hip function related tasks but not in sprint and jump related tasks or force and power related tasks which may be missed if we use scores from 1-factor model or 2-factor model.

Important Variables

	Factor loadings
1 Factor Model (Variance explained: 25%)	
F11: Physical Fitness	`sprint_20m, sprint_30m, sprint_40m, agility_left, agility_right, cmj_total, ankle_jump_left, ankle_jump_right, max_power_left, max_power_right, max_force_left, max_force_right, hamstring_left, hamstring_right,`

	Factor loadings
	hip_squeeze_0_left, hip_squeeze_0_right, hip_pull_0_left, and hip_pull_0_right
2 Factor Model (Variance explained: 24.23% and 23.79%)	
F12: Speed and Jump	sprint_20m, sprint_30m, sprint_40m, agility_left, agility_right, cmj_total, ankle_jump_left, and ankle_jump_right
F22: Knee, Hamstring, and Hip	max_power_left, max_power_right, max_force_left, max_force_right, hamstring_left, hamstring_right, hip_squeeze_0_left, hip_squeeze_0_right, hip_pull_0_left, and hip_pull_0_right
3 Factor Model (Variance explained: 23.8%, 22.5%, and 14.1%)	
F13: Speed and Jump	sprint_20m, sprint_30m, sprint_40m, agility_left, agility_right, cmj_total, ankle_jump_left, and ankle_jump_right
F23: Knee and Hamstring	max_power_left, max_power_right, max_force_left, max_force_right, hamstring_left, and hamstring_right
F33: Hip Strength	hip_squeeze_0_left, hip_squeeze_0_right, hip_pull_0_left, and hip_pull_0_right

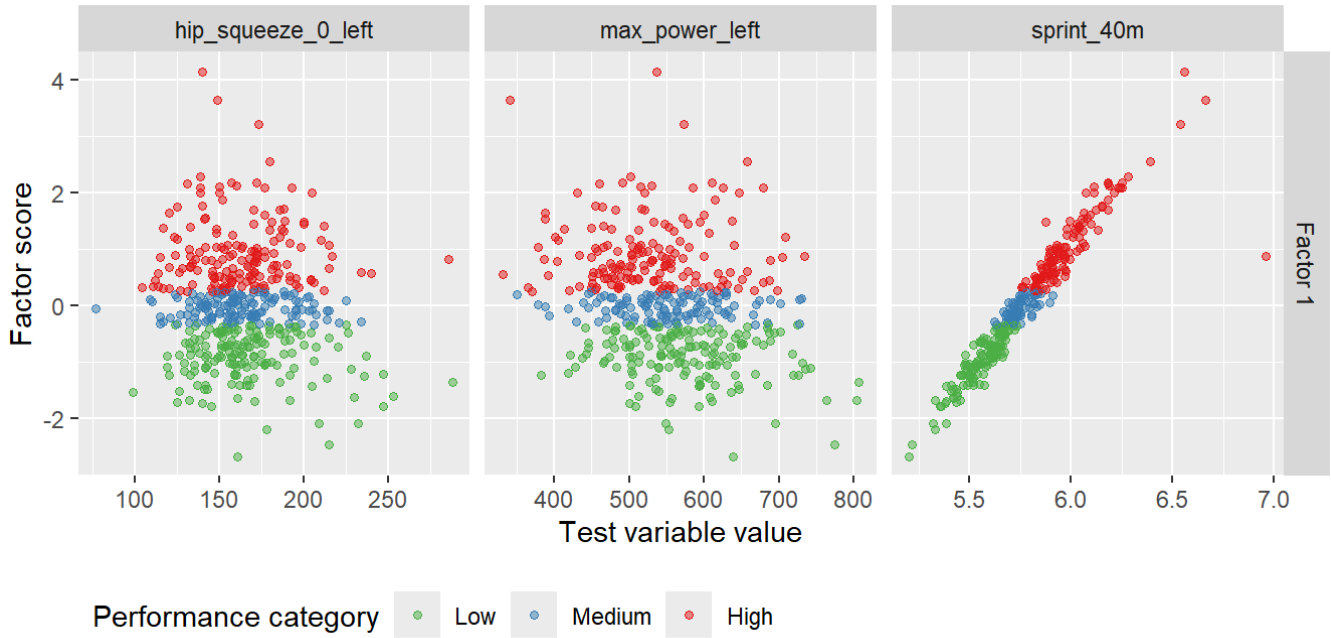
Factor, Scores, and Physical Fitness Test

When we suse 1, 2, or 3 factor model, we can calculate scores for each athlete based on the factor loadings and their test variable values. These scores represent the athlete's performance in the underlying factor domain identified by the factor analysis. For example, in the 1-factor model, the score represents the athlete's overall physical performance. In the 2-factor model, the first factor score represents the athlete's performance in the speed and power domain, while the second factor score represents the athlete's performance in the strength and force domain.

In the following plots, we can see how the scores from each factor model relate to the original test variables and the performance categories (high, medium, low) based on the scores. This helps us understand how well the factor scores capture the information from the original test variables and how they differentiate between athletes with different levels of performance.

1-factor model

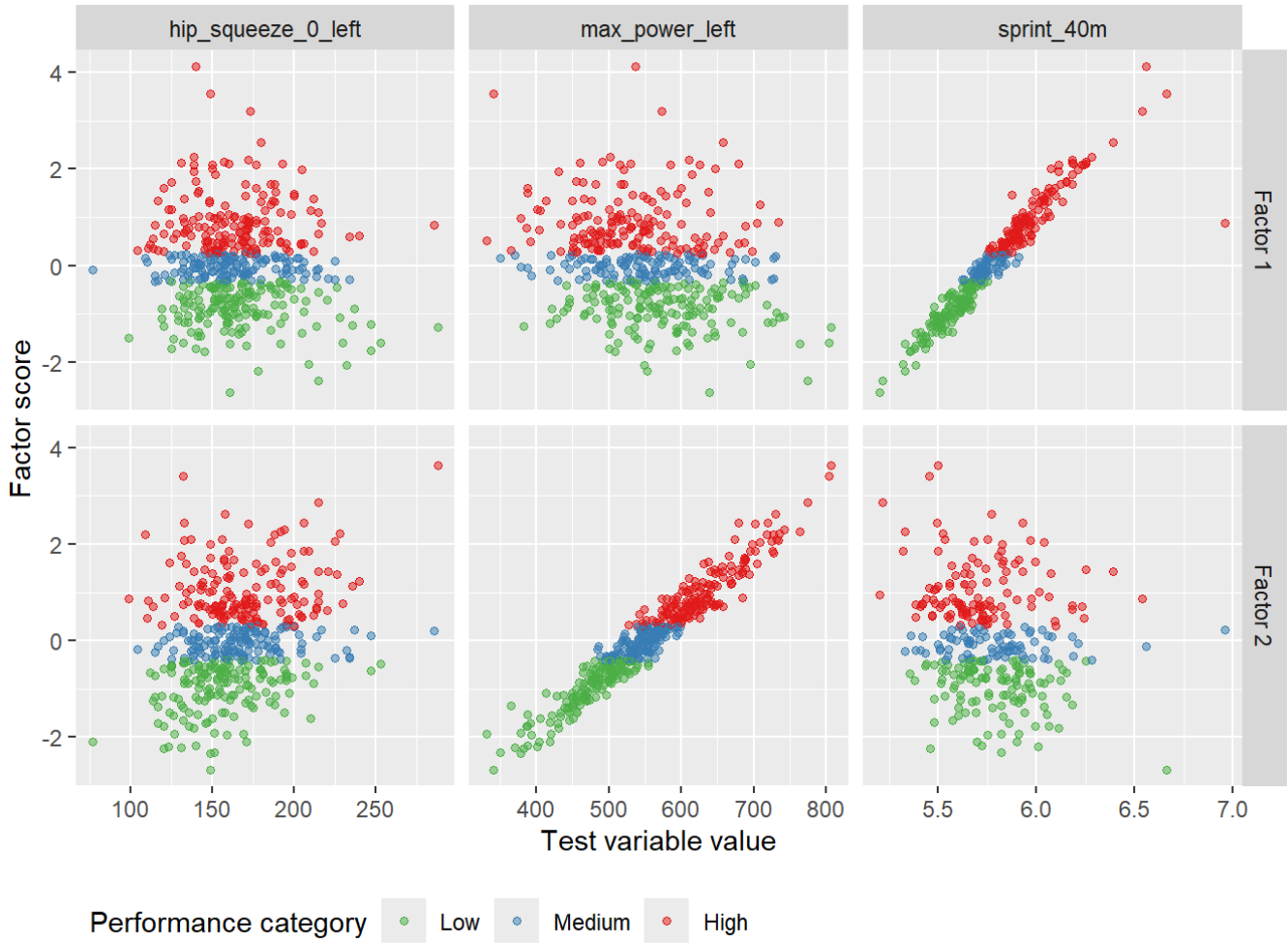
Here, we can see high correlation between scores from 1-factor model and sprint and jump related variables which is expected as these variables had the highest loadings on the factor. But for varialbes like `hip_squeeze_0_left` and `max_power_left`, the correlation with scores from 1-factor model is weaker which is also expected as these variables had lower loadings on the factor.



2-factor model

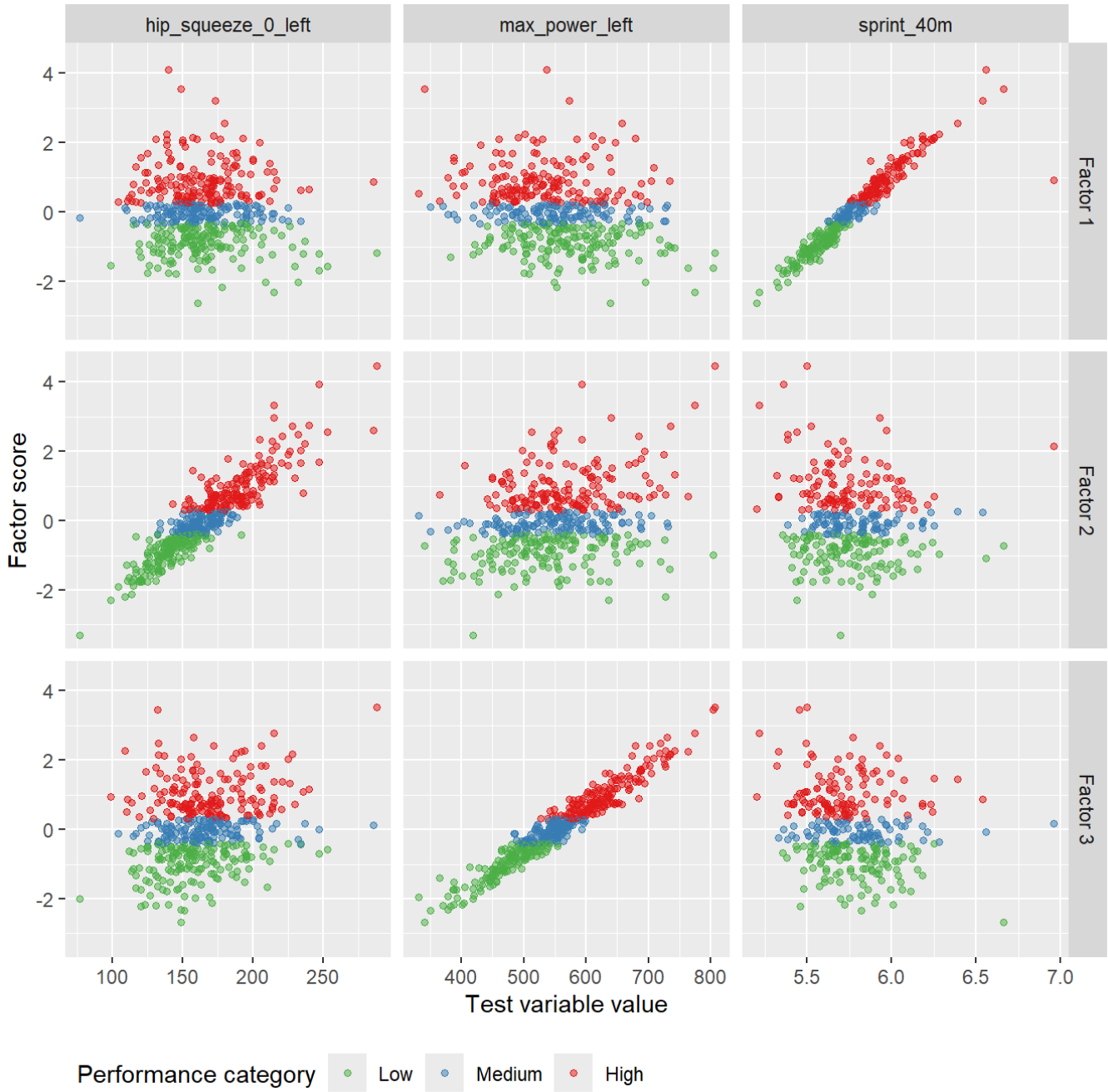
In this model, we have two factors scores, the first one has high correlation with sprint and jump related variables and the second one has high correlation with force and power related variables. This is expected as these variables had the highest loadings on their respective factors. For varialbes like `hip_squeeze_0_left`, the correlation with scores from both factors is weaker which is also expected as this variable had lower loadings on both factors. In this model, when we use the second score, we are representing the athlete's performance in the strength and force domain which is not captured by the first score from 1-factor model. This allows us to identify athletes who are good in force and power related tasks but not in sprint and jump related tasks which may be missed if we use scores from 1-factor model.

This is important when we want to analyze injuries that is more related to force and power related tasks as we can identify athletes who are at risk of such injuries based on their scores from the second factor in the 2-factor model which may not be possible if we use scores from 1-factor model.

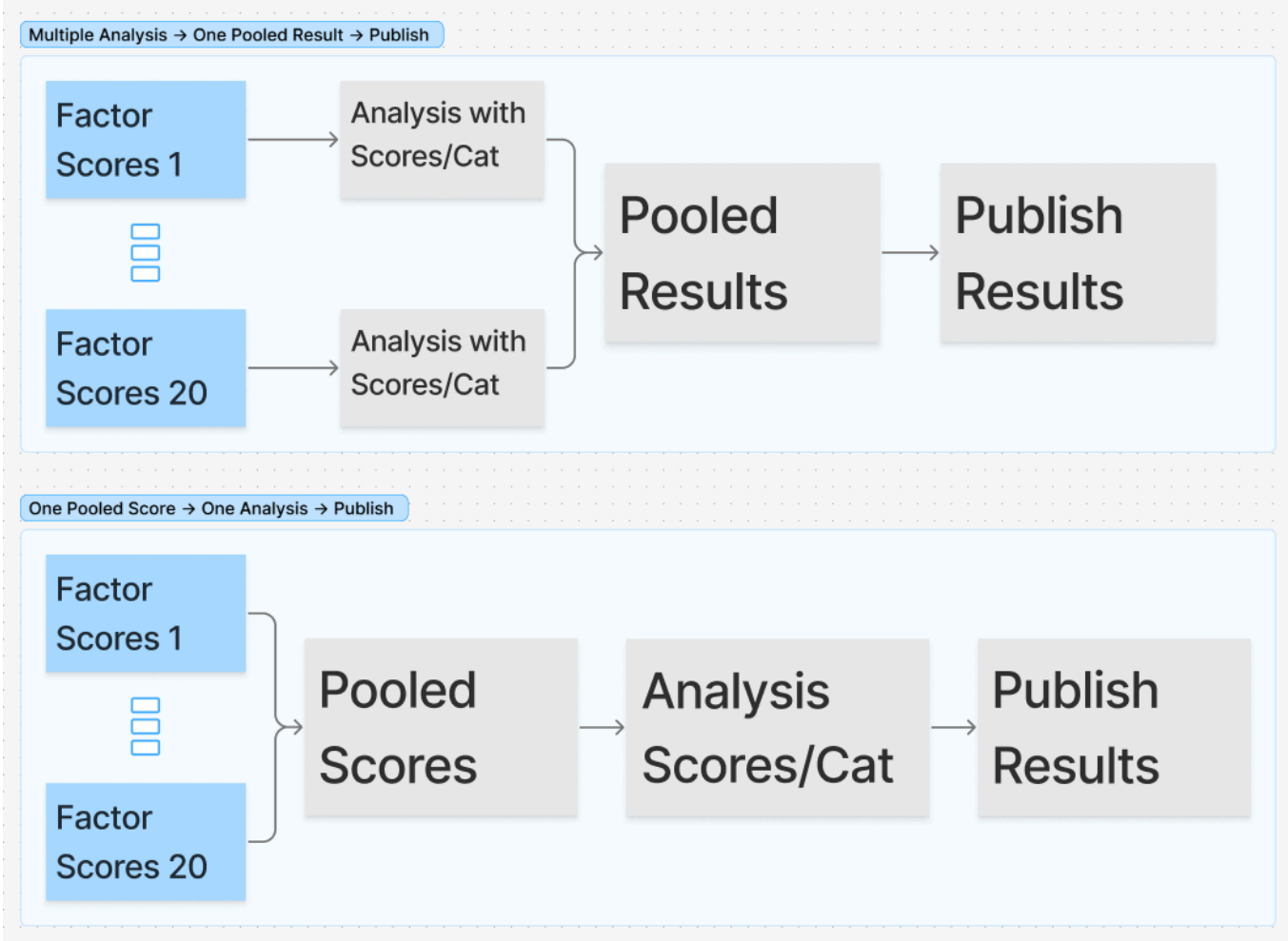


3-factor model

In the 3-factor model, there is additional dimension captured by the second factor which is related to hip function. In this model, the second factor has high correlation with hip squeeze and pull related variables. This is expected as these variables had the highest loadings on the second factor. When we use the second score from this model, we are representing the athlete's performance in the hip function domain which is not captured by the any of the scores in 1-factor model or 2-factor model. This allows us to identify athletes who are good in hip function related tasks and is important when analyzing injuries that is more related to hip function as we can identify athletes who are at risk of such injuries.



Factor analysis: 1-factor model (Physical Fitness Model)

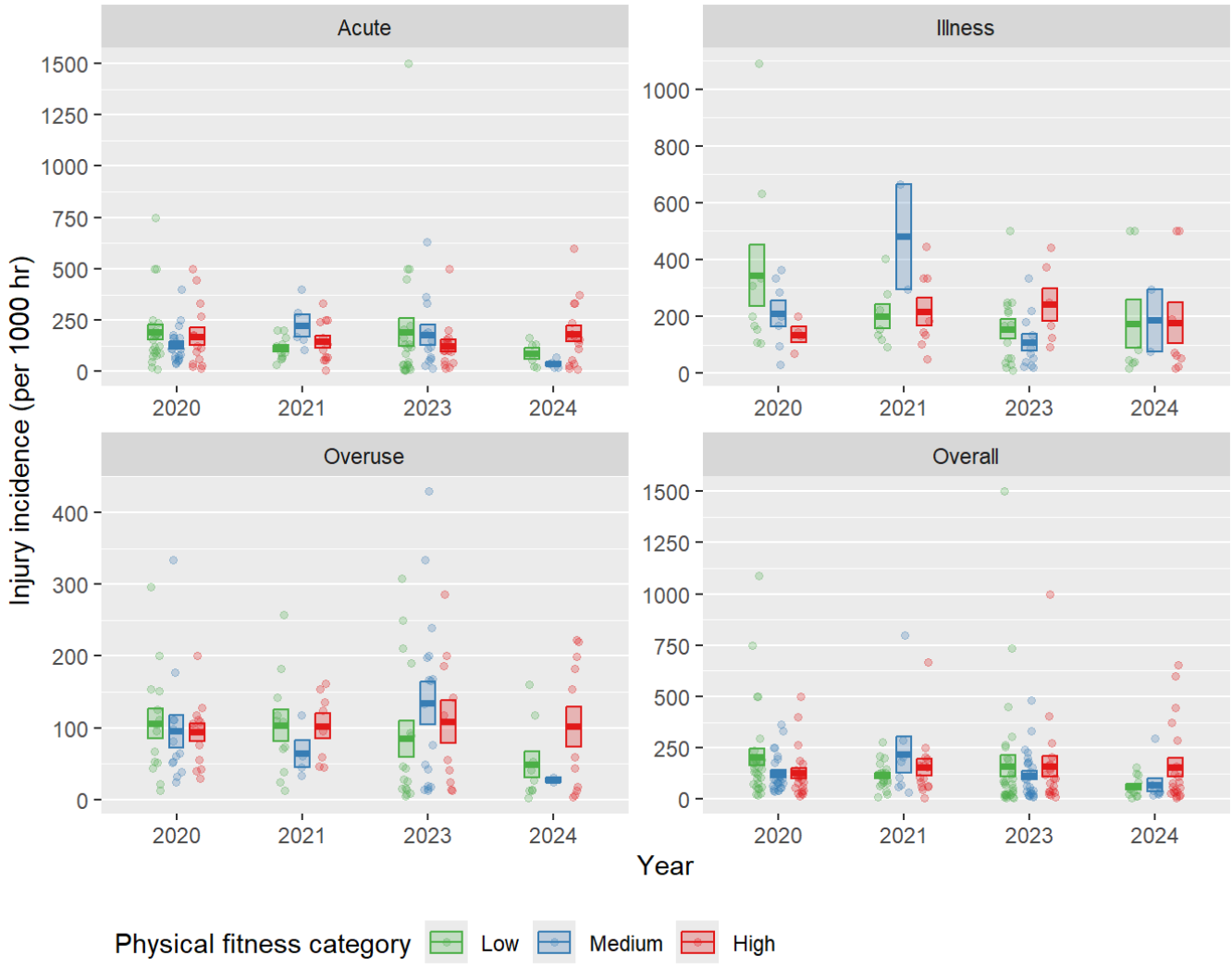


Analysis

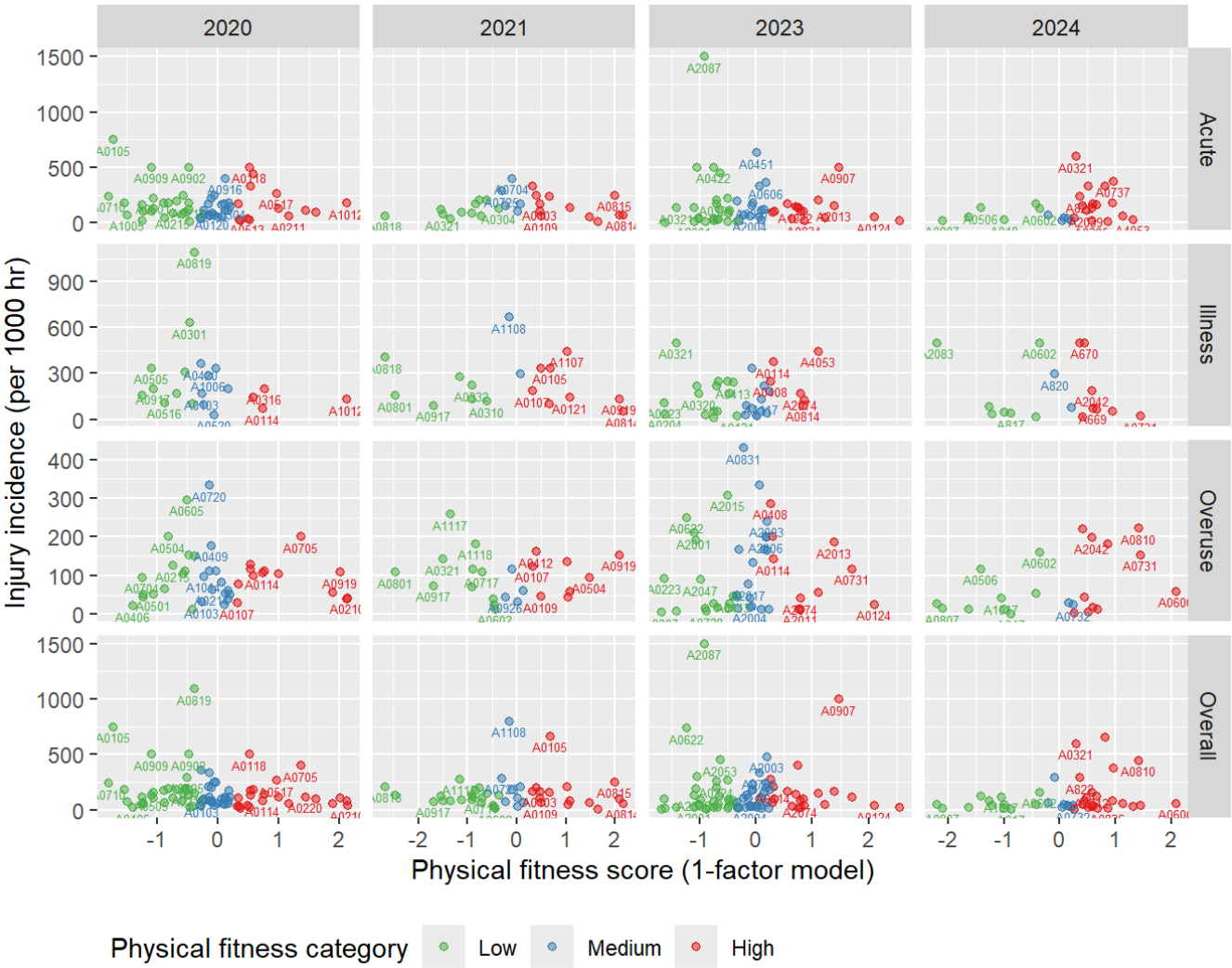
Incidence (overall and by injury type)

The following analysis is based on "one pooled score" approach illustrated in the figure above. Here, the scores from all 20 imputed datasets are pooled into one single score for each athlete and then categorized into three categories (high, medium, low) based on quantiles.

Incidence by fitness category



Incidence vs scores

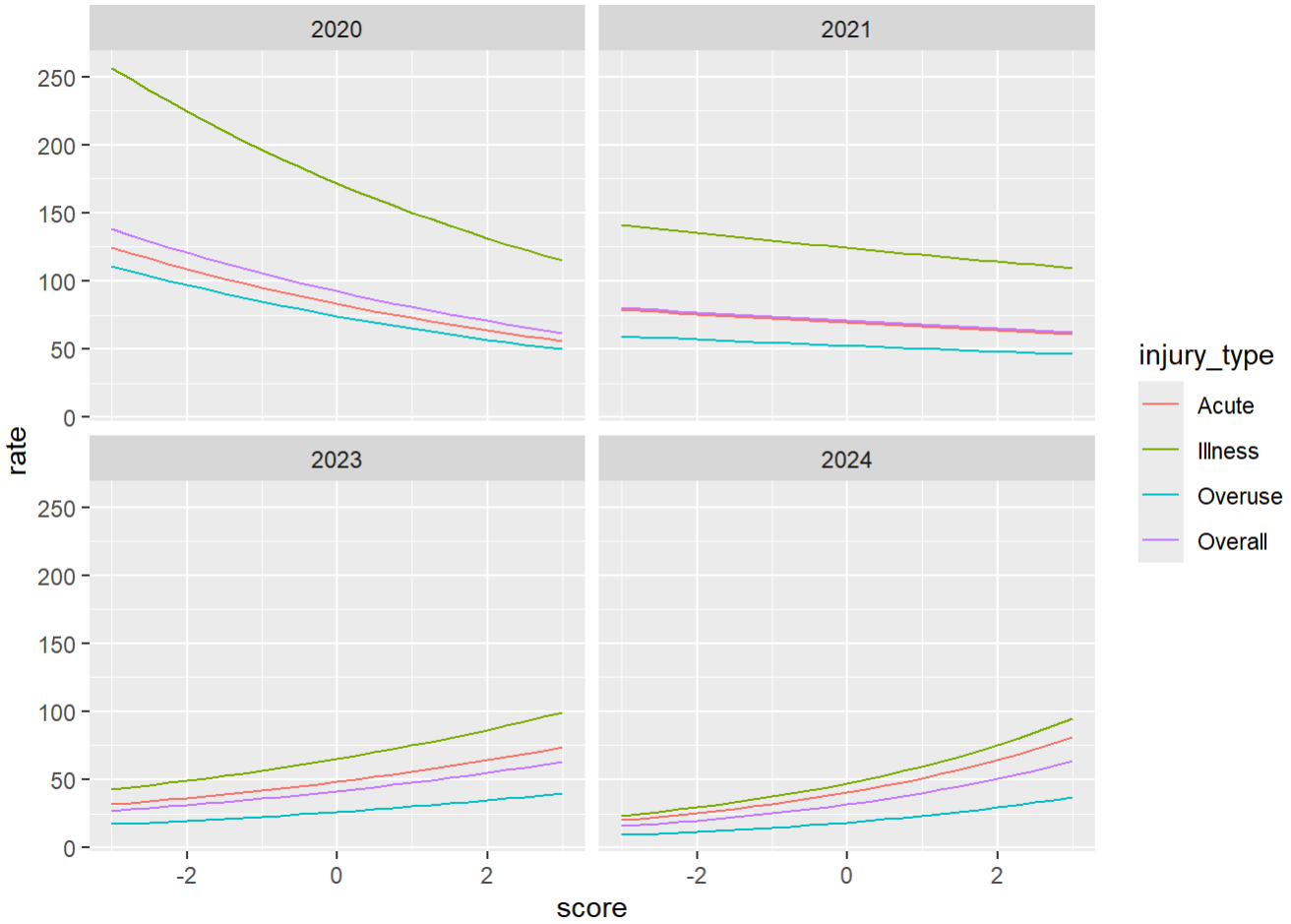


Modelling incidence using scores and fitness categories from 1-factor model

Modelling incidence vs scores from 1-factor model

Here, the continuous scores from 1-factor model are used as a predictor in the model to analyze the relationship between physical fitness and injury incidence. This allows us to see how changes in the physical fitness score are associated with changes in injury incidence, also accounting for other factors such as year and injury type.

Marginal Effects



The results shows that the injury incidence tends to decrease with physical fitness (score from 1-factor model) for overall injuries and by injury type, however this is only for the year 2020 and 2021. For the year 2023 and 2024, we observed rather reversed relationship where the injury incidence tends to increase with physical fitness (score from 1-factor model) for overall injuries and by injury type.

This is a bit unexpected and may be due to several reasons. One possible reason is that the relationship between physical fitness and injury incidence may have changed over time due to changes in training practices, injury prevention strategies, or other factors that may have influenced the relationship between physical fitness and injury incidence in these years. Another possible reason is that there may be some confounding factors that are influencing the relationship between physical fitness and injury incidence in these years which are not captured by the model.

Maybe using multiple factor model and using scores from different factors may provide more insights into the relationship between physical fitness and injury incidence as it can capture different dimensions of physical fitness which may have different relationships with injury incidence.

Model Summary

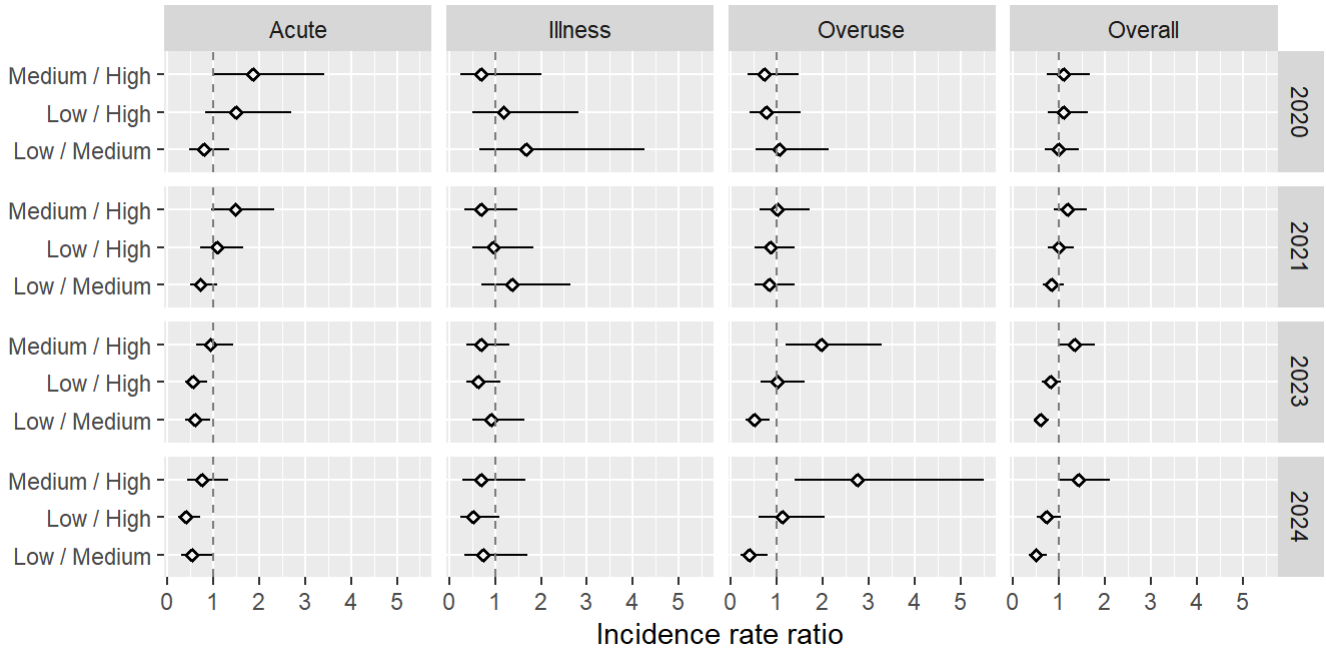
```
Call:
glm(formula = total_injury ~ year + injury_type + score + year *
    injury_type + year * score + offset(log(exposure)), family = poisson(link = "log"),
    data = model_data_fa1)

Coefficients:
                          Estimate Std. Error z value Pr(>|z|)
(Intercept)              363.05977   80.23785   4.525 6.05e-06 ***
```

```
year                      -0.18096    0.03968  -4.560 5.11e-06 ***
injury_typeIllness       287.65328  143.30426   2.007  0.04472 *
injury_typeOveruse       337.31120  120.08333   2.809  0.00497 **
injury_typeOverall       174.94371   95.10854   1.839  0.06585 .
score                   -185.34867   43.01430  -4.309 1.64e-05 ***
year:injury_typeIllness   -0.14204    0.07087  -2.004  0.04503 *
year:injury_typeOveruse   -0.16704    0.05939  -2.813  0.00491 **
year:injury_typeOverall   -0.08655    0.04704  -1.840  0.06575 .
year:score                 0.09169    0.02127   4.310 1.63e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 1495.4  on 617  degrees of freedom
Residual deviance: 1174.3  on 608  degrees of freedom
AIC: 2712

Number of Fisher Scoring iterations: 6
```

Modelling incidence vs fitness categories from 1-factor model

Here, the fitness categories (high, medium, low) based on scores from 1-factor model are used as a predictor in the model to analyze the relationship between physical fitness and injury incidence. This allows us to see how being in different fitness categories is associated with changes in injury incidence, also accounting for other factors such as year and injury type.

Marginal Effects



Model Summary

```
Call:
glm(formula = total_injury ~ year * injury_type * fitness_cat +
    offset(log(exposure)), family = poisson(link = "log"), data = model_data_fa1)

Coefficients:
                                          Estimate Std. Error z value
(Intercept)                              639.64934  136.75802   4.677
year                                      -0.31785    0.06764  -4.699
injury_typeIllness                       270.63961  220.77683   1.226
injury_typeOveruse                        95.00553  204.36102   0.465
injury_typeOverall                        67.35512  160.98266   0.418
fitness_catMedium                       -187.58354  192.19617  -0.976
fitness_catHigh                         -646.34026  197.18942  -3.278
year:injury_typeIllness                   -0.13355    0.10919  -1.223
year:injury_typeOveruse                   -0.04717    0.10107  -0.467
year:injury_typeOverall                   -0.03331    0.07962  -0.418
year:fitness_catMedium                     0.09297    0.09506   0.978
year:fitness_catHigh                       0.31977    0.09752   3.279
injury_typeIllness:fitness_catMedium     -215.51489  364.29682  -0.592
injury_typeOveruse:fitness_catMedium     -299.55906  305.19935  -0.982
injury_typeOverall:fitness_catMedium     -149.87831  231.40787  -0.648
injury_typeIllness:fitness_catHigh        232.55666  335.13629   0.694
injury_typeOveruse:fitness_catHigh        826.46108  289.47189   2.855
injury_typeOverall:fitness_catHigh        438.08154  231.95579   1.889
year:injury_typeIllness:fitness_catMedium   0.10632    0.18016   0.590
year:injury_typeOveruse:fitness_catMedium   0.14816    0.15094   0.982
year:injury_typeOverall:fitness_catMedium   0.07409    0.11445   0.647
year:injury_typeIllness:fitness_catHigh    -0.11501    0.16573  -0.694
year:injury_typeOveruse:fitness_catHigh    -0.40882    0.14315  -2.856
year:injury_typeOverall:fitness_catHigh    -0.21672    0.11471  -1.889
                                         Pr(>|z|)
(Intercept)                              2.91e-06 ***
year                                     2.62e-06 ***
```

```
injury_typeIllness                          0.22025
injury_typeOveruse                          0.64201
injury_typeOverall                          0.67565
fitness_catMedium                           0.32906
fitness_catHigh                             0.00105 **
year:injury_typeIllness                     0.22127
year:injury_typeOveruse                     0.64071
year:injury_typeOverall                     0.67568
year:fitness_catMedium                      0.32806
year:fitness_catHigh                        0.00104 **
injury_typeIllness:fitness_catMedium        0.55412
injury_typeOveruse:fitness_catMedium        0.32634
injury_typeOverall:fitness_catMedium        0.51719
injury_typeIllness:fitness_catHigh          0.48773
injury_typeOveruse:fitness_catHigh          0.00430 **
injury_typeOverall:fitness_catHigh          0.05894 .
year:injury_typeIllness:fitness_catMedium  0.55508
year:injury_typeOveruse:fitness_catMedium  0.32633
year:injury_typeOverall:fitness_catMedium  0.51741
year:injury_typeIllness:fitness_catHigh    0.48769
year:injury_typeOveruse:fitness_catHigh    0.00429 **
year:injury_typeOverall:fitness_catHigh    0.05885 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 1495.4  on 617  degrees of freedom
Residual deviance: 1141.2  on 594  degrees of freedom
AIC: 2707

Number of Fisher Scoring iterations: 6
```